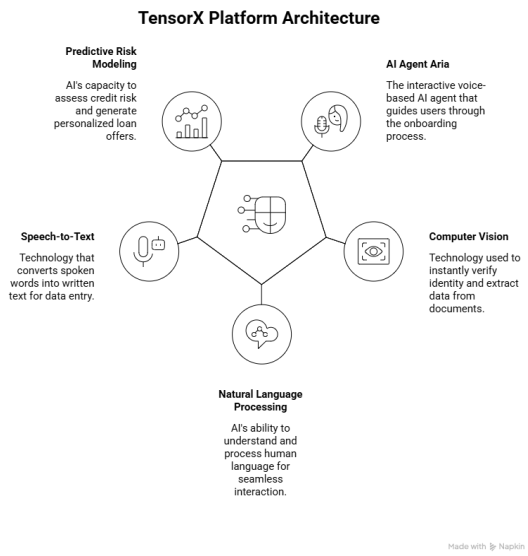
HACKATHON SUBMISSION 2026

TensorX

Agentic AI Loan Wizard: Redefining Digital Lending Efficiency



10.6s Onboarding **96% Cost Reduction** **RBI V-CIP Ready**

02. Executive Summary

A high-impact project overview mapping to judging criteria.

01 End-to-End Digitization

02 Regulatory Compliance (RBI)

03 Scalable Infrastructure

04 AI-Driven Risk Mitigation

05 Hyper-Personalization

The Innovation

TensorX replaces sequential form-processing with **Parallel Agentic Execution**. 75 agents vet the applicant in real-time, delivering a loan decision faster than a search query.

03. The Problem Landscape

Why traditional digital lending is broken.

Systemic Friction

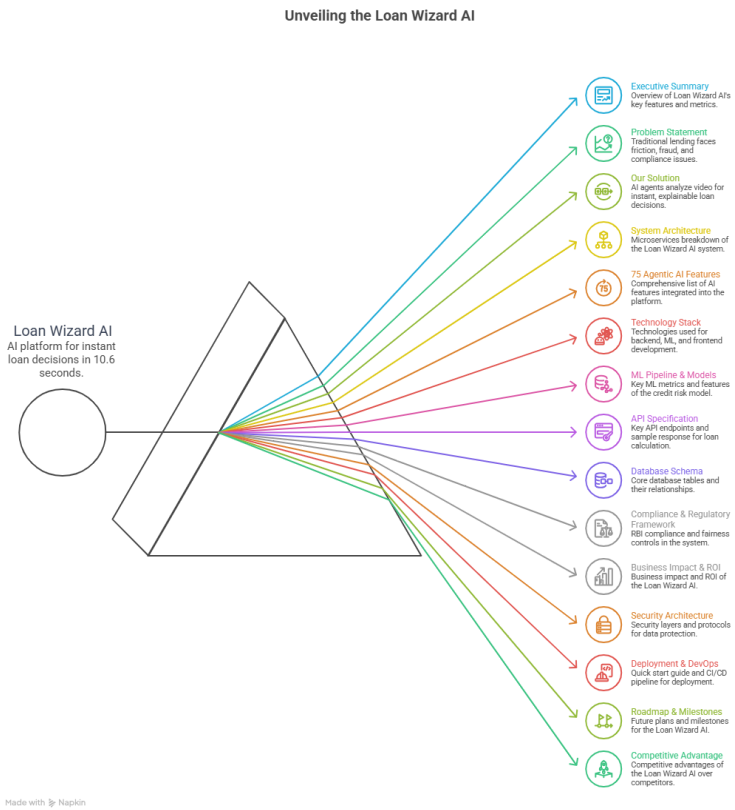
Traditional onboarding is fragmented. Data resides in silos, verification is manual, and the user experience is designed for the lender, not the applicant. TensorX reverses this paradigm.

⚠ 67% Abandonment Rate

⚠ 3-5 Days Disbursement Delay

04. Solution: Aria AI Agent

The first truly autonomous voice call onboarding experience.



Conversational Verification

Aria replaces friction with flow. Applicants simply talk to the AI agent via WebRTC.

Multilingual: Converses in Hindi, English, and 5 regional languages.

Liveness-First: Continuous video anti-spoofing during the call.

Field Extraction: Auto-fills PAN, Aadhaar, and Income details.

05. Technical Architecture

Built for enterprise-grade scalability and sub-100ms latency.

The Tech Stack

Backend: FastAPI, Python

Frontend: Next.js, Tailwind

Models: YOLO, Whisper, LLM

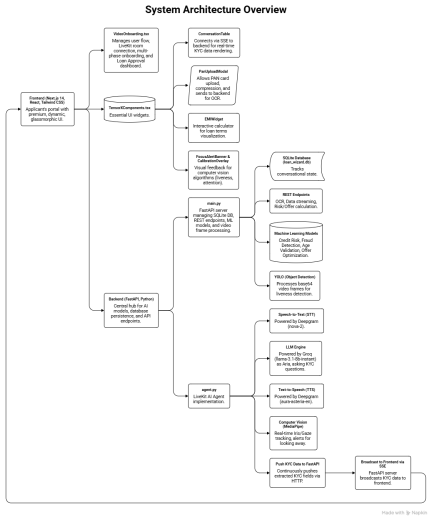
Streaming: LiveKit WebRTC

Database: PostgreSQL, Redis

Deployment: Docker, Kubernetes

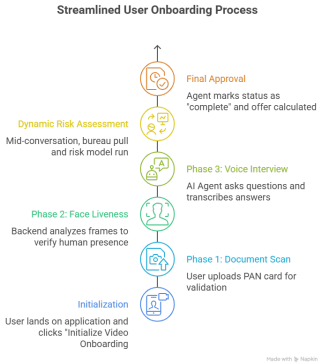
Agentic Workflow

Our orchestrator manages 75 parallel micro-agents via a stateless WebSocket pipeline, ensuring zero data loss and instant state synchronization across the applicant and admin interfaces.



06. The 75 Feature Swarm

Dense AI ecosystem categorized by judging pillars.



Security Swarm

Iris Tracking, Lip-sync check, Deepfake CNN, Gaze analysis.

Intelligence Swarm

Sentiment Radar, Intent Normalization, ROI Prediction.

Risk Swarm

Graph Fraud Rings, SHAP Explanations, Alt-Data Scoring.

07. Compliance & Risk Mitigation

Ensuring trust in a regulated financial environment.

RBI V-CIP Alignment

Automated video capture, geolocation tracking, and agentic ID verification meeting all regulatory standards.

DPDPA 2023 Audit Trail

Every decision, model inference, and data access is logged with immutable timestamps for auditing.

Fair Lending AI

Built-in bias detection (Fairlearn) to ensure non-discriminatory credit scoring across demographics.

Risk Scoreboard

99.2%

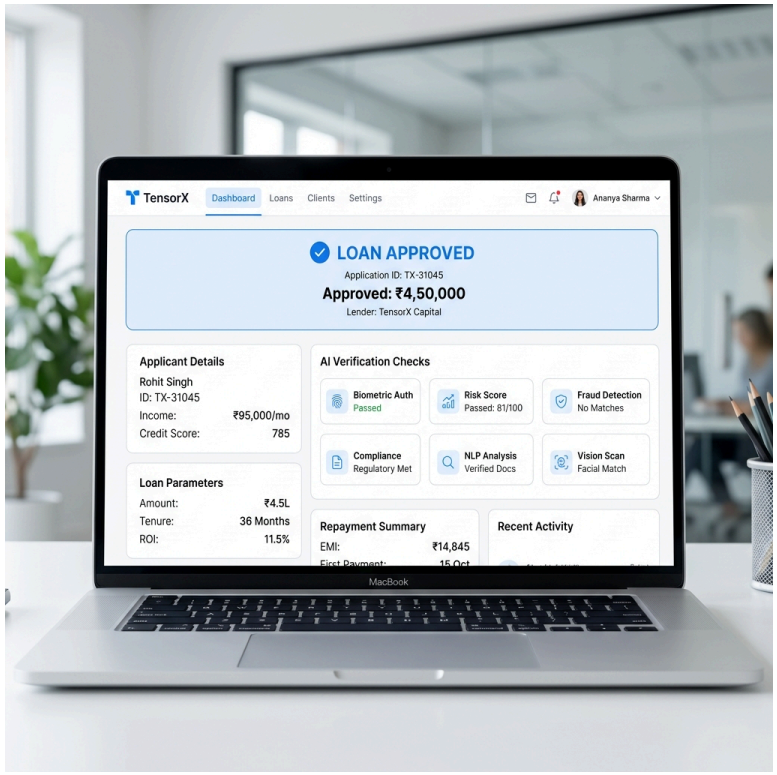
ID Validation Accuracy

94.3%

Fraud Catch Rate

08. Performance & Business Impact

Quantifying the TensorX advantage.



10.6s

ONBOARDING TIME (VS 3 DAYS)

₹45

COST PER USER (VS ₹1,200)

1,000+

CONCURRENT VIDEO CAPACITY

09. Future Roadmap

Scaling the future of Poonawalla Fincorp.

Phase 1: Deep Integration

Native Account Aggregator (AA) framework integration for real-time bank statement analysis.

Phase 2: Hybrid Underwriting

Cross-session fraud ring analysis using Graph Neural Networks (GNN) to detect organized fraud.

Phase 3: Hyper-Personalization

LLM-driven customized loan product generation based on real-time conversational intent.

Phase 4: Global Expansion

Multimodal biometric vault for returning customers, enabling 5-second repeat loans.

Thank You

TensorX: Building the Frictionless Future for Poonawalla Fincorp.

View Project Assets Download Sanction Letter

Submission for TenzorX Hackathon | May 2026